Sales Forecasting & Demand Intelligence System

Executive Business Report

Prepared for: Head of Supply Chain & Chief Financial Officer

Executive Summary

Over the past four years, our company's sales have grown steadily, but they rise and fall sharply with the seasons — November and December alone drive the biggest spikes of the year. Using our historical sales records, we built a system that predicts how much we'll sell over the next three months, flags unusual weeks that fell far outside normal patterns, and groups our products into demand categories so each one can be stocked the right way. The short version: sales are on a healthy upward path, the next three months should follow the usual pre-holiday build-up, a small number of past weeks stand out as true anomalies worth understanding, and roughly a quarter of our product lines need a different stocking approach than the rest. This report lays out what we found and what we recommend doing about it.

Key Findings from Data Exploration & Forecasting

What the historical data shows

- **Category performance:** Technology is our strongest revenue driver, generating more total sales than Furniture or Office Supplies over the full four-year period.
- **Regional trends:** Growth has been most consistent in the West region — while every region has grown, West's year-over-year growth has been the steadiest, making it a reliable bet for expansion investment.
- **Fulfillment speed:** Shipping takes about 4 days on average from order to dispatch, and this is consistent across all regions — logistics performance is not a current bottleneck.
- **Seasonality:** Sales spike predictably every November and December across all four years, consistent with holiday shopping patterns. Early Q1 (January–February) is consistently the slowest period.

How well can we predict future sales?

We tested three different forecasting approaches side by side and measured each one's accuracy against six months of actual, known sales it hadn't seen before:

Model	Avg. Error (MAE)	Error Spread (RMSE)	% Error (MAPE)	Recommended
SARIMA (statistical)	$15,448	$18,566	18.3%	Backup
Prophet (industry-standard)	$14,501	$19,156	17.8%	**Primary ✓**
XGBoost (machine learning)	$22,818	$26,384	27.0%	Not used

Prophet is the most accurate of the three, with the lowest average error and lowest percentage error, so it is used as our primary production model. SARIMA performs nearly as well and serves as a cross-check. XGBoost consistently underestimates upcoming peak months and is not recommended on its own for multi-month-ahead forecasts.

3-Month Sales Forecast

Based on our recommended model (Prophet), here is what we expect over the next three months, along with a realistic range to account for normal uncertainty:

Month	Expected Sales	Likely Low	Likely High
Month 1	$42,548	~$34,000	~$51,000
Month 2	$33,310	~$26,000	~$41,000
Month 3	$80,305	~$68,000	~$93,000

In plain terms: the next two months look like typical pre-holiday months, with Month 2 dipping slightly before Month 3 shows a sharp seasonal jump — consistent with the November/December surge we've seen in every prior year. Procurement and staffing should be scaled up ahead of Month 3, not Month 1.

Top 3 Anomalies Detected & Likely Causes

We ran two independent anomaly-detection methods against four years of weekly sales — one that scans the whole history for extreme highs/lows, and one that watches for sudden shifts relative to the last two months. The following stand out as the highest-confidence, most business-relevant anomalies:

- **1. Late November / Mid-December Spikes —** Sales surged far beyond the normal range in the weeks surrounding Black Friday and Cyber Monday, and again in mid-December. Likely cause: seasonal holiday promotions and year-end corporate bulk purchasing — expected and healthy, but worth over-staffing and over-stocking for deliberately each year.
- **2. End-of-Quarter Spike (Late March) —** A sharp, short-lived jump in sales appeared just before the end of a fiscal quarter (late March). Likely cause: business customers rushing to spend remaining quarterly budgets before it expires — a recurring B2B pattern rather than a data error.

- **3. Post-Holiday Trough (January/February) —** Sales dropped well below the expected range in the weeks immediately following the December peak. Likely cause: the predictable 'post-holiday hangover' — consumer spending pulls back and corporate procurement pauses for annual budget resets in January/February.

Only 2 of the flagged weeks were caught by both detection methods at once — these represent the highest-confidence, truly unusual events, while the rest reflect normal (if extreme) seasonal rhythm rather than one-off shocks.

Product Demand Segmentation & Stocking Strategy

We grouped all product sub-categories into four demand clusters based on sales volume, growth, volatility, and average order value. Each cluster needs a different stocking approach:

- **Cluster: Low Volume, High Volatility —** Small, unpredictable sellers (e.g., Fasteners, Labels, Art supplies). Keep minimal stock on hand and order on-demand from suppliers with fast turnaround — holding large inventory here ties up capital for little benefit.
- **Cluster: Growing Demand, Premium Value —** Products with strong upward momentum and high per-order value (e.g., Appliances, Chairs, Tables). Increase safety stock by 15–20% ahead of peak quarters so we don't lose sales to stockouts as demand accelerates.
- **Cluster: High Volume, Stable Demand —** Our steady, high-volume sellers (e.g., Paper, Binders, Storage, Phones). These are predictable enough for automated reordering — set fixed reorder points and buy in bulk to get better freight rates.
- **Cluster: Low Volume, Declining Demand —** Slow or declining sellers (e.g., Copiers, Machines, Supplies). Trim inventory commitments here and consider shifting fulfillment to a drop-ship arrangement with a third-party vendor rather than holding warehouse space.

Business Recommendations

- **1. Adopt Prophet as the standard forecasting tool for monthly procurement planning.** Data shows Prophet's forecast beats the next-best alternative (SARIMA) by roughly $950 in average monthly error and beats a machine-learning approach (XGBoost) by over 9 percentage points in accuracy — a meaningful difference when scaled across monthly procurement decisions.
- **2. Direct incremental investment toward West-region Technology stock ahead of peak season.** The West region shows the most consistent multi-year growth of any territory, and Technology is already our top-revenue category — the data supports prioritizing capital and warehouse space toward West-region Technology fulfillment ahead of the Month 3 seasonal surge.
- **3. Right-size inventory for underperforming product clusters.** A quarter of our product sub-categories fall into the declining or highly volatile clusters. Continuing to stock these at the same level as our core sellers ties up capital unnecessarily — shifting them to on-demand or drop-ship fulfillment frees up warehouse space and cash for our growing segments.

Risk & Limitation

This system's forecasts are built entirely on four years of historical patterns. It assumes the future will broadly resemble the past — normal seasonal cycles, similar customer behavior, and no major shocks. It cannot anticipate one-off disruptions such as a new competitor, a supply shortage, a pricing change, or an unexpected economic shift. The forecast should be treated as a well-informed starting point for planning, not a guarantee — and should be revisited monthly as new sales data comes in, rather than relied on as a static, one-time prediction.